Lösung: Integration durch Substitution

a) $\int \frac{1}{3x+2}\,dx$ mit $z = 3x+2$ $\left(\frac{dz}{dx} = 3\right)$

$= \int \frac{1}{z}\,\frac{dz}{3}$

$= \frac{1}{3}\int \frac{dz}{z} = \frac{1}{3}\cdot \ln(z) + C = \underline{\underline{\frac{1}{3}\ln(3x+2) + C}}$

b) $\int \frac{1}{x\cdot \ln(x)}\,dx$ mit $z = \ln(x)$ $\left(\frac{dz}{dx} = 1/x\right)$

$= \int \frac{1}{\cancel{x}\cdot z}\,\cancel{x}\cdot dz = \int \frac{dz}{z} = \ln(z) + C = \underline{\underline{\ln(\ln(x)) + C}}$

c) $\int e^{3x+2}\,dx$ mit $z = 3x+2$ $\frac{dz}{dx} = 3$

$= \int e^z\,\frac{dz}{3} = \frac{1}{3}\int e^z dz = \frac{e^z}{3} + C = \underline{\underline{\frac{e^{3x+2}}{3} + C}}$

d) $\int_0^1 \frac{-6x}{(4-3x^2)^2}\,dx$ mit $z = 4-3x^2$ $\left(\frac{dz}{dx} = -6x\right)$

$= \int_{z(0)}^{z(1)} \frac{\cancel{-6x}}{z^2}\,\frac{dz}{\cancel{-6x}} = \int_4^1 \frac{dz}{z^2} = \int_1^4 \frac{-1}{z^2}\,dz = \left[\frac{1}{z}\right]_1^4$

$= \frac{1}{4} - 1 = \underline{\underline{-\frac{3}{4}}}$

e)

$$\int \tan(x)\,dx = \int \frac{\sin(x)}{\cos(x)}\,dx \qquad \text{mit } z = \cos(x) \quad \left(\frac{dz}{dx} = -\sin(x)\right)$$

$$= \int \frac{\sin(x)}{z}\,\frac{dz}{(-\sin(x))} = -\int \frac{dz}{z}$$

$$= -\ln(z) + C = \underline{\underline{-\ln(\cos(x)) + C}}$$

f)

$$\int \sqrt{1-x}\,dx = \ldots \qquad \text{mit } z = 1-x \quad \left(\frac{dz}{dx} = -1\right)$$

$$= \int \sqrt{z}\,(-dz) = -\int \sqrt{z}\,dz = -\int z^{1/2}\,dz = -\frac{1}{3/2}\,z^{3/2} + C$$

$$= -\frac{2}{3}\,(1-x)^{3/2} + C = \underline{\underline{-\frac{2}{3}\left(\sqrt{1-x}\right)^3 + C}}$$